Shreya Komarabattini

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Education

Purdue University
Bachelor of Science in Computer Science **05/2026**

Skills

Programming Languages: TypeScript, JavaScript, Python, Java, C++, SQL, HTML, CSS
Frontend: React.js, Next.js, Vite, Tailwind CSS, WCAG/ARIA Accessibility
Backend & APIs: Node.js, Express, FastAPI, REST APIs, Microservices, PostgreSQL, MongoDB, Redis, Docker, AWS
Testing & Tooling: Cypress, Supertest, Jest, CI/CD, GitHub Actions, Git, Agile, Scrum, Kanban
AI & Dev Tools: Claude Code, GitHub Copilot, LangChain, RAG, Prompt Engineering, OpenAI SDK

Projects

Moodle - Real-Time Multiplayer Drawing & Guessing Game *(React.js, TypeScript, Node.js, Socket.IO, PostgreSQL, Redis, Docker)* **05/2026**
Link https://moodl3.com/

- Designed **distributed backend services and WebSocket (Socket.IO) channels** in Node.js for real-time multiplayer features (rooms, scoring, chat, state sync) across concurrent users; integrated **PostgreSQL and Redis** for relational persistence and distributed session caching.
- Improved game reliability with **Redis-backed room persistence**, automatic player reconnection, **server failover**, and fallback AI behavior when external services are unavailable.
- Wrote **Cypress end-to-end tests** and maintained automated test coverage; participated in code review workflows and Agile Scrum sprint cycles throughout development.

ProtoPlay - Interactive Engineering Learning Platform *(Next.js, TypeScript, Tailwind CSS, shadcn/ui)* **01/2026**
Link https://protoplay.vercel.app

- Built an **interactive platform** in **Next.js and TypeScript** with 20+ challenge-based modules teaching distributed systems concepts (caching, load balancing, microservices, and scaling) through visual simulations and hands-on exercises.
- Achieved a **100 Lighthouse Performance score** across core learning pages through targeted rendering optimization, strategic code splitting, and component-level performance tuning.
- Designed the full system architecture and UI component library using **Tailwind CSS and shadcn/ui**, balancing technical depth with an intuitive, high-quality user experience.

Traffic Sign Recognition Android App *(Kotlin, C++, JNI, OpenCV, ONNX Runtime, YOLOv8)* **05/2026**
Link https://github.com/rishigxsh/Traffic-Sign-Recognition-App

- Built a **real-time Android mobile application** that detects and classifies traffic signs through a live camera feed, delivering bounding-box overlays and prioritized **Text-to-Speech** voice alerts for a clear, hands-free driver experience.
- Achieved **15-30 FPS at sub-200 ms latency** fully offline using **ONNX Runtime** and a C++ JNI backend with OpenCV preprocessing; designed alert logic to prevent overlapping announcements under continuous inference load.
- Deployed on-device **YOLOv8 inference** via a native C++ layer bridged through JNI, optimizing the mobile rendering pipeline to sustain reliable performance across constrained hardware.

Experience

IT Lab Consultant, *Purdue University* **08/2023 – 05/2026**

- Re-imaged, configured, and maintained 100+ lab computers, ensuring consistent OS, software installations, and security updates.
- Maintained **99% lab uptime** by resolving **1,500+** hardware and software tickets, triaging incidents by priority, and implementing proactive monitoring and preventive maintenance to minimize disruptions for students and faculty.
- Performed routine diagnostics, printer maintenance, and **troubleshooting** of technical equipment.
- **Collaborated** with IT staff while independently managing lab systems and planning lab events.

Undergraduate Researcher: Do We Really Know How to Use Graphs Effectively | Dr. Beomjin Kim **09/2024 – 02/2025**

- Researched how non-domain experts interpret categorical visualizations: bar, pie, grouped bar, and stacked bar charts.
- Designed user studies to evaluate visualization clarity and reduce misinterpretation by non-expert audiences.
- Formulated evidence-based guidelines for selecting appropriate chart types to improve data comprehension.
- Presented findings at the 28th Annual Student Research and Creative Endeavors Symposium.

Teaching Assistant, Senior Capstone Project, *Purdue University* **02/2025 – 04/2025**

- Assisted in evaluating capstone project reports and presentations.
- Leveraged Python scripts to support data processing and review pipelines.

CS Summer Camp Assistant, *Purdue University* **06/2025 – 07/2025**

- Represented the Computer Science department during a high-school summer camp program.
- Guided high school students through coding fundamentals via hands-on programming exercises and project-based activities.
- Conducted research on prompt engineering and evaluated student capstone posters using AI-based tools.

Leadership & Presentations

- Active member of the Student Government Organization, organizing and supporting campus events.
- Presented research findings at the Love Data 2025 workshop at Purdue University, discussing data visualization practices and effective interpretation of categorical data.